

TED[15] 5214
(REVISION-2015)

Reg. No.....
Signature.....

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
MODEL QUESTION PAPER: OPTICAL INSTRUMENTATION

[Time : 3 hours]

(Maximum marks : 100)

PART - A
(Maximum marks: 10)

Marks

I Answer all questions in one or two sentences. Each question carries 2 marks.

1. State Law of reflection.
2. Define multi mode graded indexed fibre.
3. List any four applications of liquid dye laser.
4. Define Laser drilling.
5. Define holography.

(5x2=10)

PART - B
(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Explain dispersion with a neat diagram.
2. Explain the principles of optical fibre communication.
3. With the help of a neat diagram explain the working of Argon laser.
4. List the applications of Laser in Instrumentation.
5. Explain interference.
6. Explain passive optical fibre sensors.
7. Explain thermal equilibrium

(5x6=30)

PART - C
(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT - I

- III.a. Explain phase difference and path difference . (8)
- b. Explain Polarization. . (7)

OR

- IV.a. Explain Diffraction . (8)
- b. Explain the laws of Polarization . (7)

UNIT – II

- V.a. Explain mode of operations of optical fibres . (8)
- b. Draw and explain the working of optical fibre torque sensor. (7)

OR

- VI.a. Explain LED drive circuits. (8)
- b. Draw and explain the working of optical fibre strain sensor. (7)

UNIT – III

- VII.a. Explain with a neat diagram the construction and working of CO₂ laser. (8)
- b. Explain population inversion. (7)

OR

- VIII.a. Explain the working of liquid dye laser with a energy -level diagram (8)
- b. Explain stimulated emission. (7)

UNIT – IV

- IX.a. Explain Laser cutting. (8)
- b. Explain medical applications of laser . (7)

OR

- X. a. Explain scientific applications of laser. (8)
- X. b Explain laser Doppler velocity meter . (7)